

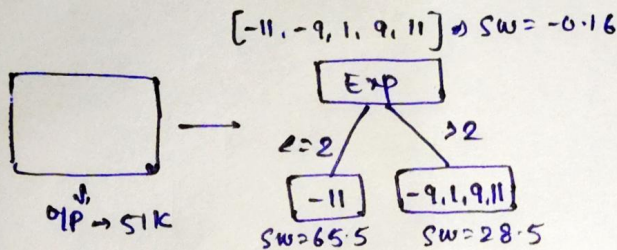
• Xg Boost Regressor ML Algorithm:

Dataset (Regressor)

Exp	Gap	Salary	R_1	$\hat{y}$	R_2
2	Yes	40K	-11	50	-10
2.5	Yes	42K	-9	50	-8
→ 3	No	52K	1	51.5	0.5
4	No	60K	9	51.5	8.5
4.5	Yes	62K	11	52.1	9.9
		$\approx 51K$			

Steps

1. Create a Base Model
- ii. Residual computation
- iii. Construct DTL using $\{x_i, R_i\}$
- iv. Calculate Gain



$$\text{Similarity weight} = \frac{\sum (\text{Residual})^2}{\text{No. of Residuals} + \lambda}$$

$\lambda \uparrow SW \downarrow$ Hyperparameters

$$SW(\text{left child}) = \frac{121}{1+1} = \frac{121}{2} = 65.5$$

$$SW(\text{right child}) = \frac{(-9+1+9+11)^2}{4+1} = 28.5$$

Calculate Gain:

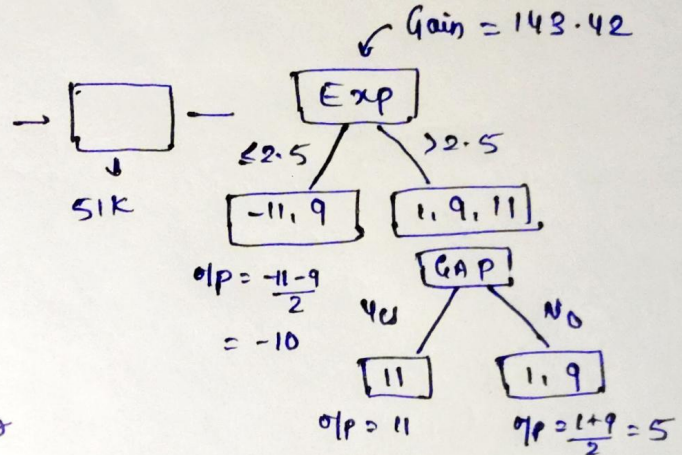
$$\text{Gain} = 65.5 + 28.5 - 0.16 = 98.84$$

$$\text{XG Boost Classifier} = \text{Base learner} + \alpha_1 (DT_1) + \alpha_2 (DT_2) + \dots + \alpha_n (DT_n)$$

$$\begin{aligned} \text{XG Boost Regressor} &= 51K + 0.1(5) \\ &= 51 + 0.5 = 51.5 \end{aligned}$$

$$\text{Similarity weight} = \frac{\sum (\text{Residuals})^2}{\text{No. of Residuals} + \lambda}$$

↓
Regressor



$$[51 + (0.1)(-10)] = 51 - 1 = 50$$

$$[51 + (0.1)(5)] = 51.5$$

$$[51 + (0.1)(11)] = 52.1$$

$$\text{Similarity weight} = \frac{\sum (\text{Residuals})^2}{\sum P_i (1 - P_i) + \lambda}$$

↓
Classifier